CHARMOULA WMA9ROU4 · TRACK 1, PROBLEM 1 — "PROCESS OR TRANSMIT?"

ASTRA: Adaptive Satellite Task and Resource Allocator

A decision engine for onboard Earth-observation satellites
— process, queue, transmit, or drop.

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THE DECISION IN ORBIT

Short contact makes every payload decision count.



Process · Queue · Transmit · Drop

WHAT ASTRA IMPLEMENTS

One simulator, four competing policies.

Payload → Scheduler → Process / Queue / Downlink / Drop

Battery · Thermal · Storage · Ground contact

- Transmit-all · Greedy-edge · Threshold heuristic · MaskablePPO

Implemented & reproducible today · FaaS and safety guard = future design

MODEL AND EVALUATION

A paired, multi-orbit synthetic benchmark.

4
policies

5
shared seeds

4
orbits per episode

Mean · SD · 95% CI

Synthetic simulation — not flight data

RESULTS

Completion and delivered value tell different stories.

2.86%
Heuristic — highest delivered-utility ratio

5.17%
Greedy-edge — most tasks completed



5 paired episodes · synthetic workload

WHAT THE PROTOTYPE HAS NOT SOLVED

PPO failed; the thermal threshold is not a hard cap.

PPO CHECKPOINT

0

tasks completed · single training seed

Lower energy = inactivity, not efficiency.

THERMAL TRACE

52.48 °C

peak · threshold 50 °C



WHAT'S NEXT

ASTRA gives a reproducible way to evaluate these trade-offs.

Calibrate → Add safety guard → Compare → Validate

- **Why it matters:** spend scarce contact time and power on the most valuable data.
- Next: thermal model · Lyapunov + oracle · more seeds
- All results are synthetic, not flight-validated.
- Thank you — Team Charmoula wMa9rou4 · Track 1, Problem 1